

Full Stack Development with MERN

Project Documentation format

1. Introduction

- **Project Title:** HouseRental
- **Team Member:** Self Project.

2. Project Overview

- **Purpose:** This House Rentals a web platofrom to book or buy a properly's from owners, easy access to the users, simple register with your username, mail id and role.

- **Features :**

- 1.User registration and login
- 2.Role-based authentication (Admin/ Renter/ Owner)
- 3.All Properties in Admin Side.
- 4.Admin dashboard
- 5.View all Bookings by admin
- 6.Secure API communication
- 7.MongoDB database storage

3. Architecture

- **Frontend:** Vite + React ,with tailwindCSS.
- **Backend:** Node.js and Express.js, with MVC pattern.
- **Database:** MongoDB.

4. Setup Instructions

- **Prerequisites:** List software dependencies (e.g., Node.js, MongoDB, React, Express and vite+ React)..
- **Installation:** git clone <https://github.com/Yuvaraj-kurri3/HouseRental>
Cd client: npm install

To run : npm run dev

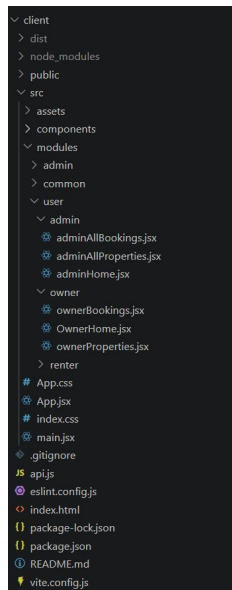
Cd server: npm install

To run: npm run dev

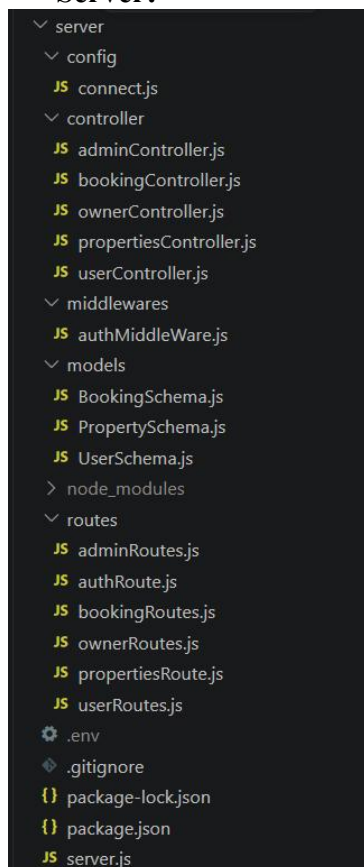
Add Env vairables: MONDB_URL,JWT_SECRET.

5. Folder Structure

Client:



- **Server:**



6. Running the Application

- Provide commands to start the frontend and backend servers locally.
 - **Frontend:** `cd client : npm run dev` in the client directory.
 - **Backend:** `cd server: npm run dev` in the server directory.

7. API Documentation

- Register: : /api/users/register
- Login: /api/users/login
- Authentication : /api/auth/protected
- Add Property: /api/owner/newProperty
- Get property:/api/owner/myProperties/:OwnerId
- Delete Property: /api/owner/deleteProperty/:PropertyId
- Update Property: /api/owner/updateProperty/:propertyId
- Book Property: /api/booking/properties/book/:propertyId

◆ PostMan Collection Link : [ClickHere](#)

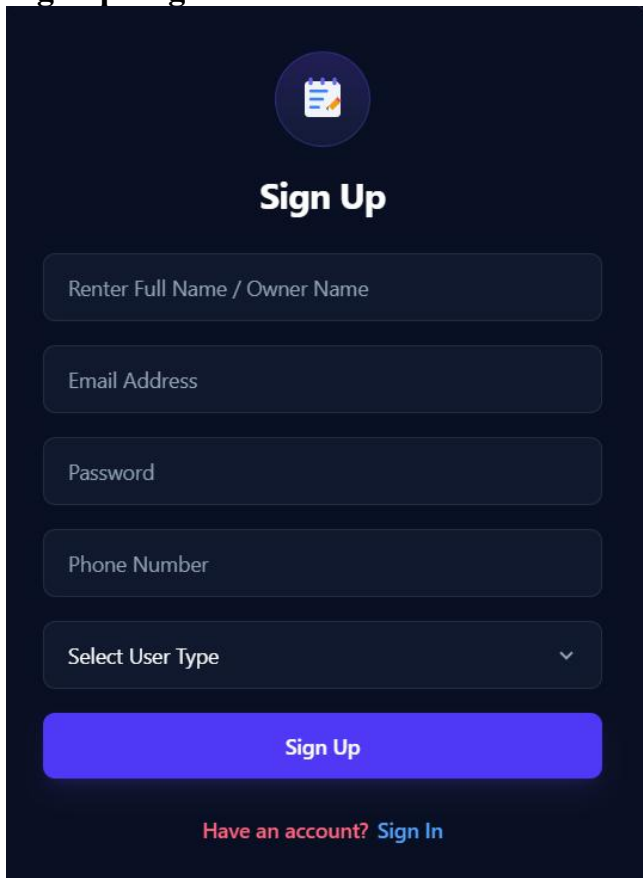
8. Authentication

Authentication: Login with registered mailID and Password.

Authorization: (Roll Based Access)RBA, it navigate to dashboard based on Role's.and each user password is encrypted and after login it generates jwt Token. And JWT expires time is 1Hour (1h)

9. User Interface

SignUp Page:



Sign Up

Renter Full Name / Owner Name

Email Address

Password

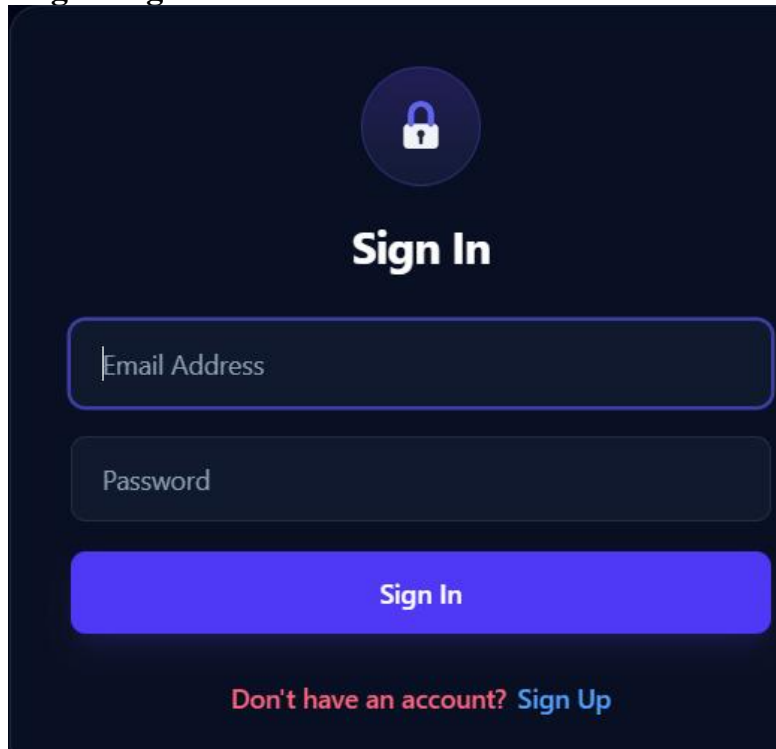
Phone Number

Select User Type ▼

Sign Up

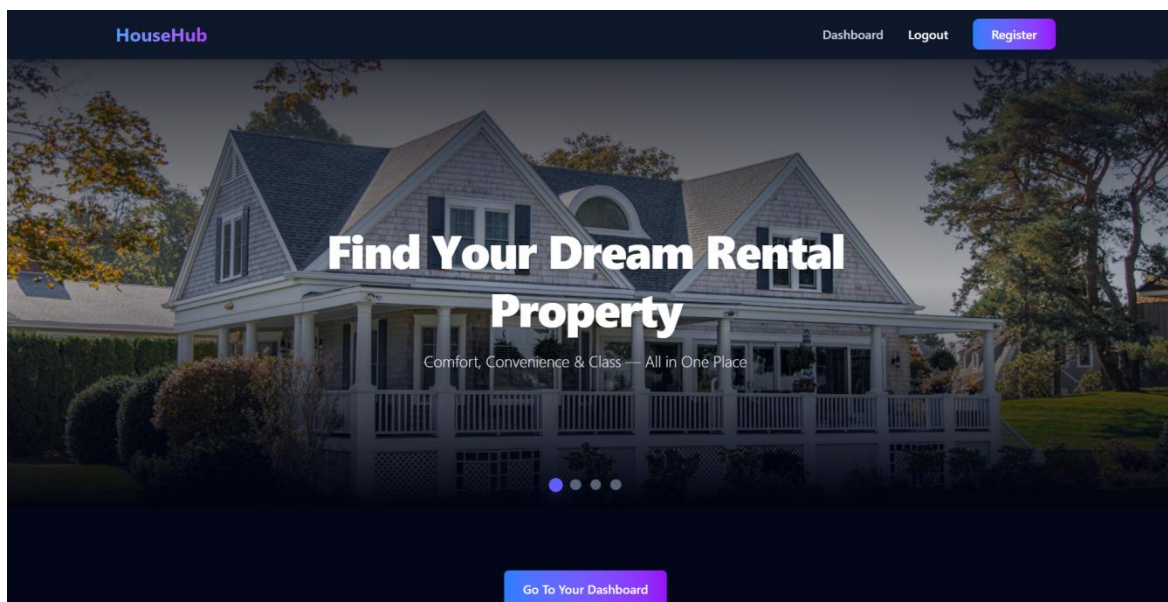
Have an account? [Sign In](#)

Login Page:



A screenshot of a login page with a dark blue background. At the top center is a circular icon containing a white padlock. Below it, the text "Sign In" is displayed in a large, bold, white font. Underneath are two input fields: the first is labeled "Email Address" and the second is labeled "Password". Below these fields is a large, rounded rectangular button with a blue gradient, labeled "Sign In" in white. At the bottom, there is a link that reads "Don't have an account? Sign Up", where "Don't have an account?" is in red and "Sign Up" is in blue.

HomePage:



OwnerHome(Add Property):

HouseHub

Dashboard

Logout

Register

Add Property

All Properties

All Bookings

Add New Property

Property Type

Residential

Property Ad Type

Rent

Property Full Address

Address

Property Images

enter img

Owner Contact No.

Contact number

Property Amount

0

Availability

Available

Additional Details for the Property

Add any details here...

Add Property

All Properties:

HouseHub

Dashboard

Logout

Register

Add Property

All Properties

All Bookings

ID	Type	Ad Type	Address	Contact	Amount	Status	Actions
6a2e79bec7c10127d10f65a5	Villa	Rent	opp Anurag University, Jodimetla.	9052940332	₹0.1L	Available	Edit Delete

Owner Bookings:

HouseHub

DashboardLogoutRegister

Add PropertyAll PropertiesAll Bookings



No Properties Found
Start by adding a new property to see them listed here.

Admin(All Users):

HouseHub

LogoutRegister

All UsersAll PropertiesAll Bookings

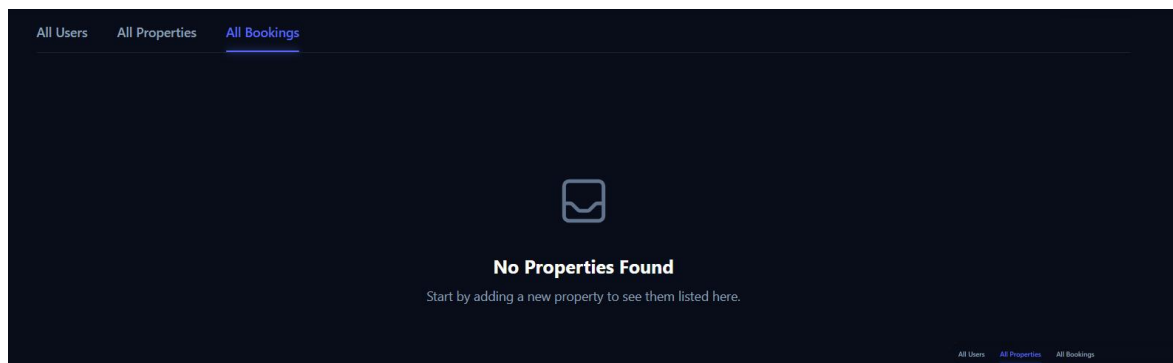
User ID	User Name	Email	Type	Presnet Permission	Update To	Delete
6a282a3fcd7c8c87d459cbcc	Yuvaraj	Kurriyuvaraj13@Gmail.Com	Renter	NA	NA	Delete
6a2e797cc7c10127d10f65a4	Yuva kurri	Yuvakurri@Gmail.Com	Owner	Granted	To UnGrant	Delete

Admin(All Properties):

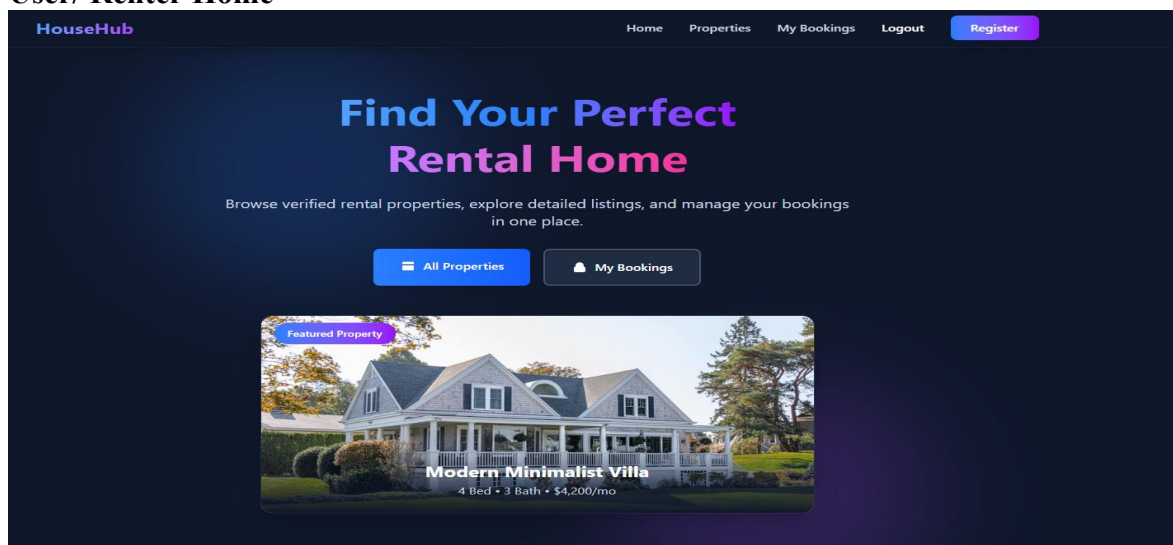
All UsersAll PropertiesAll Bookings

Property ID	Owner ID	Property Type	Property Ad Type	Propert Address	Owner Contact	Property Amount
6a2e79bec7c10127d10f65a5	6a2e797cc7c10127d10f65a4	Villa	Rent	Opp Anurag University, Jodimetla.	9052940332	12220

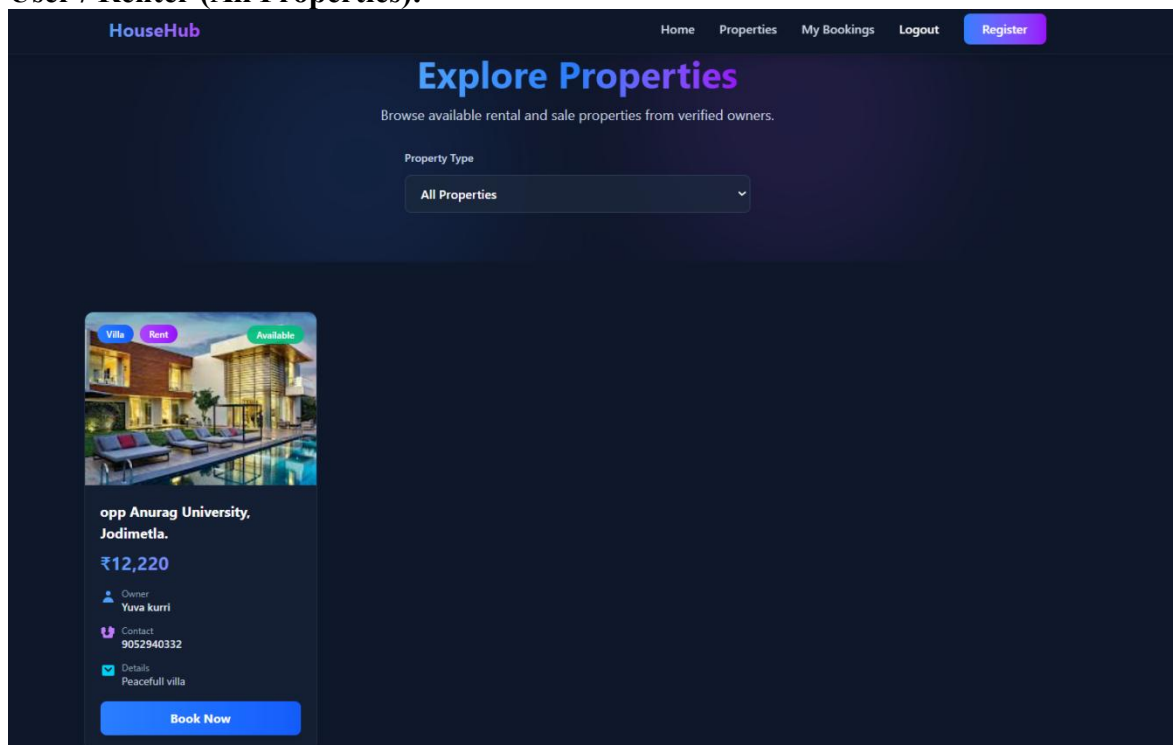
Admin(All Bookings of users/ renters):



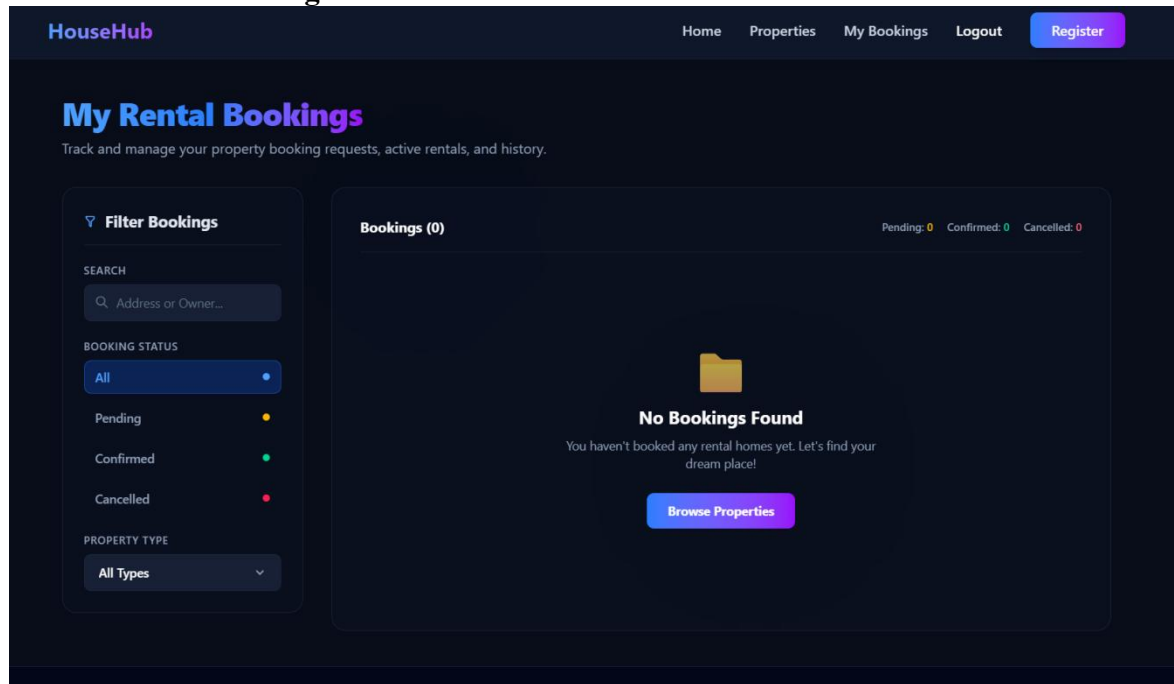
User/ Renter Home



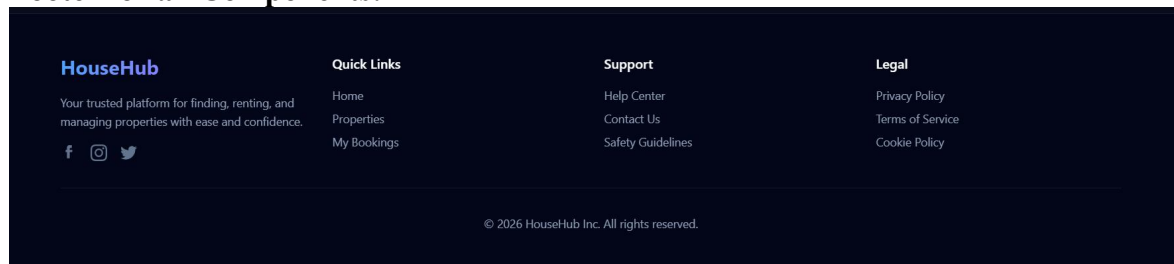
User / Renter (All Properties):



User / Renter Bookings:



Footer for all Components:



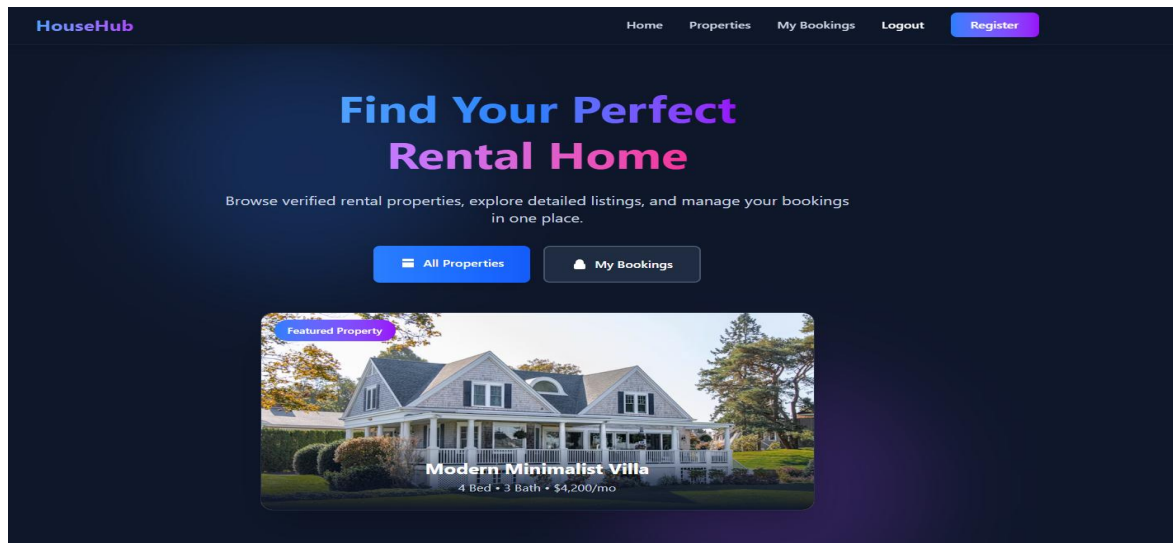
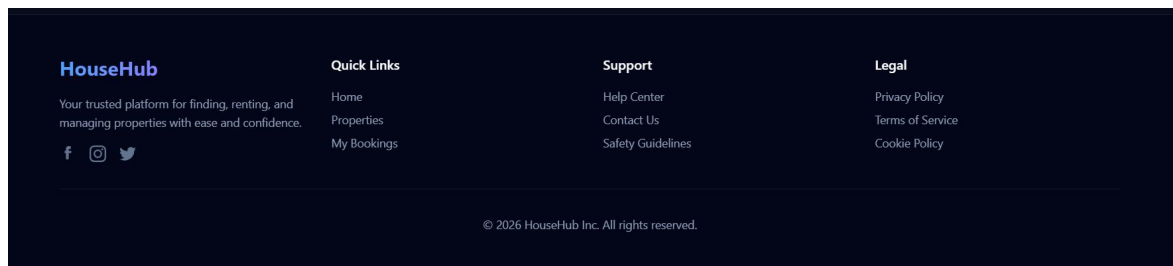
10. Testing

>Frontend Testing

- Form validation
- Page navigation
- UI testing

>Backend Testing

- Authentication testing
- Database operations testing
- Tools:
- Postman
- Browser Developer Tools
- Screenshots or Demo
- API testing using Postman



All Users All Properties All Bookings						
Property ID	Owner ID	Property Type	Property Ad Type	Property Address	Owner Contact	Property Amount
6a2e79bec7c10127d10f65a5	6a2e797cc7c10127d10f65a4	villa	Rent	Opp Anurag University, Jodimetla.	9052940332	12220

11. Known Issues

- Payment gateway not implemented
- Email notifications are not added
- Appointment conflict checking can be improved
- Doctor availability management can be enhanced

12. Future Enhancements

- Integrate with Oauth.
- Enabling Paymnet.
- Implement AI for doubts.
- Gps and payment tracking.